Revision for Test 1: Measurement

Perimeter + Area.

* FYI test 1 will also include our next topic right-angled triangles.

Review set 2A

1 Convert:

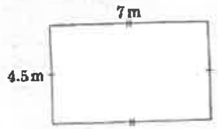
a 4.31 m to cm   b 24 300 m to km   c 5.08 m to mm.

2 Blake enjoys racing 18-foot skiffs. Write the length of his boat in:

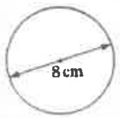
a yards   b metres.

3 Find the perimeter of:

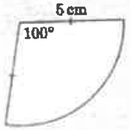
a



b



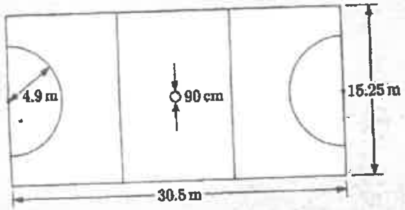
c



4 A netball court has the dimensions shown.

a Find the perimeter of the court.

b Find the total length of all the marked lines.



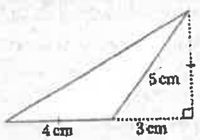
5 Convert:

a 6 m^2 to cm^2 b 1500 ha to km^2 c 4.1 acres to m^2.

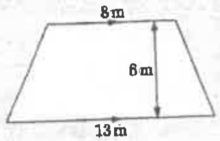
6 Fiona purchased 400 m^2 of plastic sheeting. She needs to cut it into rectangles of area 500 cm^2. This can be done with no waste. How many rectangles can Fiona make?

7 Find the area of:

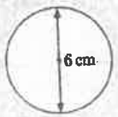
a



b



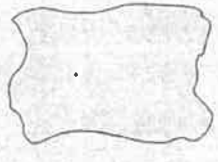
c



8 A factory fire is emitting toxic fumes, so all homes within a 500 m radius of the factory are to be evacuated. Find the area of the evacuation zone in:

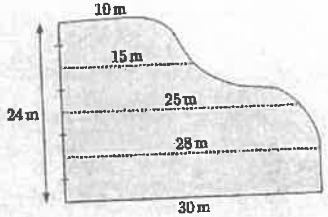
a m^2 b hectares.

9 Estimate the area of this irregular shape by approximating it with a rectangle.



PRINTABLE DIAGRAM

10 Use Simpson's rule to estimate the area of this swimming pool.



Review set 2B

1 Convert:

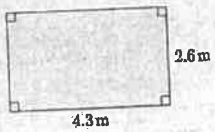
a 8 yards to feet   b 3 inches to cm

c 12 m to yards.

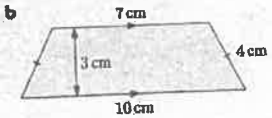
2 Betty works in a canteen. She uses 40 cm of a roll of cling wrap to wrap each sandwich. How many sandwiches can she wrap with a 60 m roll of cling wrap?

3 Find the: i perimeter   ii area of each shape.

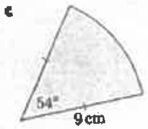
a



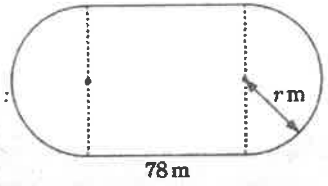
b



c



4 An athletics track has perimeter 400 m and straights which are 78 m long.
Find the radius r of the circular 'bend'.



5 Convert:

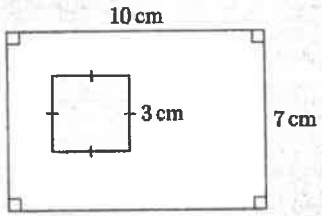
a $0.8\ m^2$ to cm^2 **b** 17.1 ha to m^2 **c** $9.02\ km^2$ to ha.

6 A traditional quarter-acre block of land is subdivided into two house blocks of equal size. Find the area of each house block in square metres.

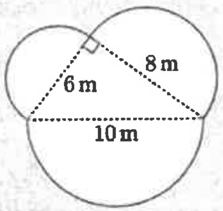
7 Answer the **Opening Problem** on page 36.

8 Find the shaded area:

a



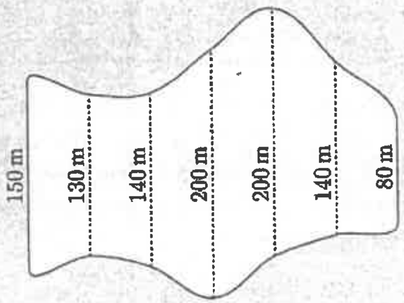
b



9 Estimate the area of this irregular shape by approximating it with a triangle.

PRINTABLE DIAGRAM

10 For the rural property shown alongside, each region is 50 m wide.
Use Simpson's rule to estimate the area of the property.



Answers:

REVIEW SET 2A

1 **a** 431 cm **b** 24.3 km **c** 5080 mm
2 **a** 6 yards **b** 5.4864 m
3 **a** 23 m **b** ≈ 25.1 cm **c** ≈ 18.7 cm
4 **a** 91.5 m **b** ≈ 156 m
5 **a** $60\,000\ cm^2$ **b** $15\ km^2$ **c** $\approx 16\,600\ m^2$
6 8000 rectangles **7** **a** $8\ cm^2$ **b** $63\ m^2$ **c** $\approx 28.3\ cm^2$
8 **a** $\approx 785\,000\ m^2$ **b** ≈ 78.5 ha
9 $\approx 4 \times 2.5 \approx 10\ cm^2$ **10** $\approx 524\ m^2$

REVIEW SET 2B

1 **a** 24 feet **b** 7.62 cm **c** ≈ 13.1 yards
2 150 sandwiches
3 **a** **i** 13.8 m **ii** $11.18\ m^2$
b **i** 25 cm **ii** $25.5\ cm^2$
c **i** ≈ 26.5 cm **ii** $\approx 38.2\ cm^2$
4 $r \approx 38.8$ **5** **a** $8000\ cm^2$ **b** $171\,000\ m^2$ **c** 902 ha
6 $\approx 506\ m^2$
7 **a** $94.5\ m^2$ **b** 1.4175 kg
c The back lawn has an irregular shape so we cannot find its exact area.
d Gordon could use a rectangle to approximate the irregular shape, or he could use Simpson's rule, in order to estimate the area of his back lawn.
8 **a** $61\ cm^2$ **b** $\approx 103\ m^2$ **9** $\approx 4.5\ cm^2$
10 $\approx 46\,500\ m^2$

Revision for Test 1: Measurement -

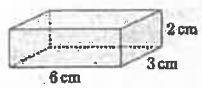
Surface Area, Volume + Capacity.

* FYI test 1 will also include our next topic right-angled triangles.

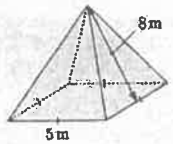
Review set 5A

1 Find the surface area of these solids:

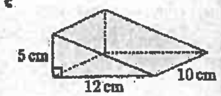
a



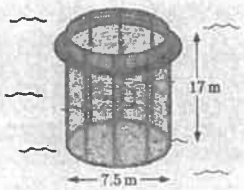
b



c

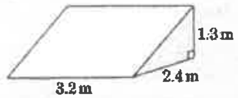


2 A fish farm has six netted cylindrical cages which are open at the top. The cylinders have depth 17 m and diameter 7.5 m. Find the total area of netting in the cages.

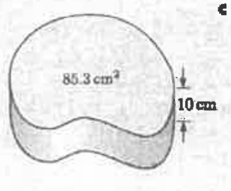


3 Calculate, to 3 significant figures, the volume of:

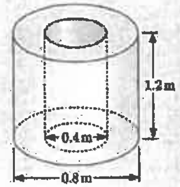
a



b



c



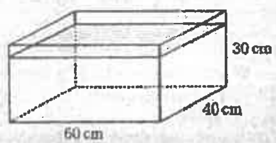
4 A plastic beach ball has radius 27 cm. Find its:

a surface area **b** volume.

5 Convert:

a 2.6 L to mL **b** 37 800 mL to kL **c** 89 m^3 to L

6 Jonathon has bought a fish tank with the dimensions shown. He fills the tank with water up to 5 cm from the top of the tank.
To make the water safe for fish, Jonathon must add 1 drop of dechlorinator for every 5 litres of water.



a Find the capacity of the tank.

b How many litres of water did Jonathon put in the tank?

c How many drops of dechlorinator should Jonathon add to the tank?

7 Convert:

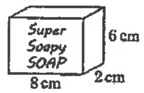
a 7.2 kg to g **b** 475 mg to g **c** 69.7 kg to t

8 An elevator has a maximum load limit of 1.2 tonnes. Assuming an average weight of 80 kg per person, how many people can safely ride in the elevator at a time?

9 Find the density of:

a a pebble with mass 19 g and volume 7 cm^3

b the bar of soap alongside which has mass 100 g.



10 Kate has $12 000 that she would like to invest in gold. She finds that the current price of gold is $1700 per ounce.

a How many 1 ounce gold coins can Kate buy?

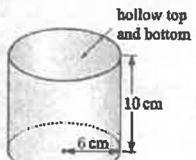
b Find the mass of Kate's coins in grams.

c The density of gold is 19.3 g/cm^3. Find the volume of Kate's coins.

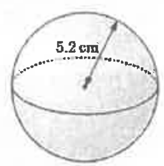
Review set 5B

1 Find, to 1 decimal place, the outer surface area of:

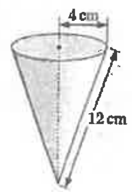
a



b



c



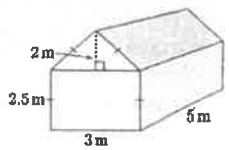
2 Convert:

a 1.2 m^3 to cm^3 **b** 13 850 mm^3 to cm^3 **c** 0.0419 cm^3 to mm^3

3 A tool shed with the dimensions shown is to be painted with two coats of zinc-alum. Each litre of zinc-alum covers 5 m^2 and costs $8.25. It must be purchased in whole litres.

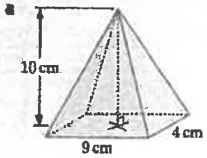
a Find the area to be painted including the roof.

b Find the total cost of the zinc-alum.

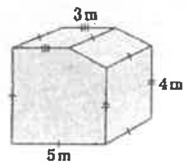


4 Find the volume of:

a



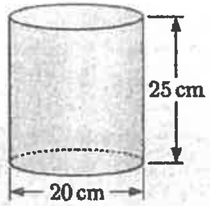
b



c

5 Find the capacity of this bucket in:

a millilitres **b** litres.



6 A rectangular shed has a roof of length 12 m and width 5.5 m. Rainfall from the roof runs into a cylindrical tank with base diameter 4.35 m. If 15.4 mm of rain falls, how many millimetres does the water level in the tank rise?

7 Convert:

a 5 lb to oz **b** 80 lb to kg **c** 12 oz to g

8 The nutritional information for a breakfast cereal is shown alongside.
Nancy ate one serving of the cereal.

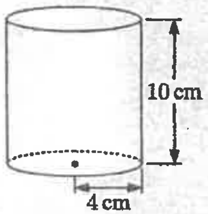
a How much sodium did she consume?

b Nancy's recommended daily sodium intake is 1.5 g. What percentage of her daily sodium intake did she just consume?

NUTRITION INFORMATION (Average)
Serving size: 33g (2 biscuits) Servings Per Pack: 42

	PER SERVE	PER 100g
Energy (Kj)	492	1490
Protein (g)	4.1	12.4
Fat, Total (g)	0.4	1.3
-Saturated Fat (g)	0.1	0.3
-Trans Fat (g)	0.0	0.0
-Polyunsaturated Fat (g)	0.2	0.8
-Monounsaturated Fat (g)	0.1	0.2
Carbohydrate, Total (g)	22.1	67.0
-Sugars (g)	1.1	3.3
Dietary Fibre (g)	3.6	11
Sodium (mg)	89	270
Potassium (mg)	112	340

9



The empty glass shown has mass 300 g. When the glass is completely filled with milk, the total mass of the milk-filled glass is 820 g.
Find the density of the milk.

10 A feed silo is made out of sheet steel using a hemisphere, a cylinder, and a cone.

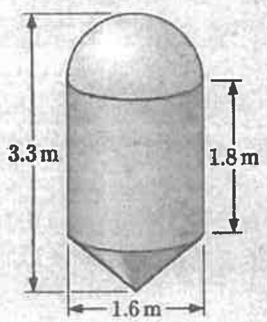
a Explain why the height of the cone must be 70 cm.

b Hence find the *slant height* of the conical section.

c Calculate the total amount of sheet steel used.

d Show that the silo can hold about 5.2 cubic metres of grain.

e The grain has density 680 kg/m^3. Find the total mass of grain in the silo when it is completely full.



Answers

REVIEW SET 5A

1 **a** 72 cm^2 **b** 105 m^2 **c** 360 cm^2
2 $\approx$ 2670 m^2
3 **a** $\approx$ 4.99 m^3 **b** 853 cm^3 **c** $\approx$ 0.452 m^3
4 **a** $\approx$ 9160 cm^2 **b** $\approx$ 82 400 cm^3
5 **a** 2600 mL **b** 0.0378 kL **c** 89 000 L
6 **a** 72 L **b** 60 L **c** 12 drops
7 **a** 7200 g **b** 0.475 g **c** 0.0697 t
8 15 people **9** **a** $\approx$ 2.71 g/cm^3 **b** $\approx$ 1.04 g/cm^3
10 **a** 7 coins **b** $\approx$ 198 g **c** $\approx$ 10.3 cm^3

REVIEW SET 5B

1 **a** $\approx$ 377.0 cm^2 **b** $\approx$ 339.8 cm^2 **c** $\approx$ 201.1 cm^2
2 **a** 1 200 000 cm^3 **b** 13.85 cm^3 **c** 41.9 mm^3
3 **a** 71 m^2 **b** $239.25
4 **a** 120 cm^3 **b** 120 m^3 **c** $\approx$ 10 300 mm^3
5 **a** $\approx$ 7850 mL **b** $\approx$ 7.85 L **6** $\approx$ 68.4 mm
7 **a** 80 oz **b** $\approx$ 36.3 kg **c** $\approx$ 340 g
8 **a** 89 mg **b** $\approx$ 5.93% **9** $\approx$ 1.03 g/cm^3
10 **a** height = 3.3 m − 1.8 m − 0.8 m = 0.7 m = 70 cm
b $\approx$ 1.06 m **c** $\approx$ 15.7 m^2
d Hint: Volume of silo
= volume of hemisphere + volume of cylinder
+ volume of cone
e $\approx$ 3510 kg